

Part 1: Theoretical Solutions

Problem 1: Geometry of Optimization and Learning Rates

(a) Derive the analytical gradient vector.

$$J(\vec{w}) = 3w_1^2 - 6w_1 + w_2^2 - 4w_2 + 8$$

$$\nabla J(\vec{w}) = \begin{bmatrix} \frac{\partial J}{\partial w_1} \\ \frac{\partial J}{\partial w_2} \end{bmatrix} \quad \frac{\partial J}{\partial w_1} = 6w_1 - 6 \quad / \quad \frac{\partial J}{\partial w_2} = 2w_2 - 4$$

$$\Rightarrow \nabla J(\vec{w}) = \begin{bmatrix} 6w_1 - 6 \\ 2w_2 - 4 \end{bmatrix}$$

(b) Find the stationary optimal point

$$\nabla J(\vec{w}) = 0 \rightarrow 6w_1 - 6 = 0 \rightarrow 6w_1 = 6 \rightarrow \boxed{w_1 = 1}$$

$$2w_2 - 4 = 0 \rightarrow 2w_2 = 4 \rightarrow \boxed{w_2 = 2}$$

$$\Rightarrow \vec{w}^* = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad H = \begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix}$$

The global minimum

(c) Perform one Gradient Descent update

$$\text{update rule: } \vec{w}^{(t+1)} = \vec{w}^{(t)} - \alpha \nabla J(\vec{w}^{(t)})$$

$$\text{Initial point: } \vec{w}^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{Gradient at the initial point:}$$

$$\nabla J(\vec{w}^{(0)}) = \begin{bmatrix} 6(0) - 6 \\ 2(0) - 4 \end{bmatrix} = \begin{bmatrix} -6 \\ -4 \end{bmatrix}$$

Candidate A: $\alpha = 0,5$

$$\vec{w}_A^{(1)} = \begin{bmatrix} 0,3 \\ 0,2 \end{bmatrix}$$

$$\vec{w}^{(1)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0,5 \begin{bmatrix} -6 \\ -4 \end{bmatrix} = \begin{bmatrix} 0 + 0,3 \\ 0 + 0,2 \end{bmatrix}$$

Candidate B: $\alpha = 0,2$

$$\vec{w}^{(1)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0,2 \begin{bmatrix} -6 \\ -4 \end{bmatrix} = \begin{bmatrix} 0 + 1,2 \\ 0 + 0,8 \end{bmatrix}$$

$$\vec{w}_B^{(1)} = \begin{bmatrix} 1,2 \\ 0,8 \end{bmatrix}$$

(d) Compute the Euclidean distances

The optimal point: $\vec{w}^* = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

Candidate A: $\| \vec{w}_A^{(1)} - \vec{w}^* \|_2 = \sqrt{(0,3-1)^2 + (0,2+2)^2}$
 $= \sqrt{(-0,7)^2 + (-1,8)^2} = \sqrt{0,49 + 3,24} = \sqrt{3,73} \quad d_A = 1,9313$

Candidate B: $\| \vec{w}_B^{(1)} - \vec{w}^* \|_2 = \sqrt{(1,2-1)^2 + (0,8-2)^2} = \sqrt{(0,2)^2 + (-1,2)^2}$
 $= \sqrt{0,04 + 1,44} = \sqrt{1,48} \quad d_B = 1,2166$

Analysis: $d_B < d_A$

Candidate B achieves faster convergence because its learning rate is larger and remains within the stable range.

The stability condition is: $0 < \alpha < \frac{2}{\lambda_{\max}}$

$$0 < \alpha < \frac{2}{6}$$

$$0 < \alpha < 0,3333$$

oscillation.

Since:

$$\alpha = 0,2 < 0,3333$$

Candidate B converges faster without oscillation

Problem 2: Multi-Feature Gradient Descent Manual Trace

Given: $X = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix} \quad \vec{y} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix} \quad \vec{w}^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad b^{(0)} = 0 \quad \alpha = 0,1 \quad m=3$

(a) Model, Cost Function, and Update Rules

Linear model $\hat{y} = X\vec{w} + b$ For each sample: $\hat{y}_i = w_1 x_{i,1} + w_2 x_{i,2} + b$

Mean Squared Error $J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i)^2$

In matrix notation: $J(\vec{w}, b) = \frac{1}{m} (\vec{X}\vec{w} + b\vec{1} - \vec{y})^T (\vec{X}\vec{w} + b\vec{1} - \vec{y})$

Define: $\vec{r} = \vec{X}\vec{w} + b\vec{1} - \vec{y}$

Gradient with Respect to Weights

$$\nabla_{\vec{w}} J = \frac{2}{m} \vec{X}^T \vec{r}$$

Gradient with Respect to Bias

$$\frac{\partial J}{\partial b} = \frac{2}{m} \sum_{i=1}^m r_i$$

Update Rules

$$\vec{w}^{(t+1)} = \vec{w}^{(t)} - \alpha \nabla_{\vec{w}} J \quad b^{(t+1)} = b^{(t)} - \alpha \frac{\partial J}{\partial b}$$

(b) Epoch 1 Initial Parameters $\vec{w}^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $b^{(0)} = 0$

Predictions $\hat{y}^{(0)} = X \vec{w}^{(0)} + b^{(0)} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
 $\hat{y}^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Residuals $\vec{r}^{(0)} = \hat{y}^{(0)} - y = \begin{bmatrix} 0 & -2 \\ 0 & -5 \\ 0 & -3 \end{bmatrix} \Rightarrow \vec{r}^{(0)} = \begin{bmatrix} -2 \\ -5 \\ -3 \end{bmatrix}$

Gradient of Weights $\nabla_{\vec{w}} J = \frac{2}{3} X^T \vec{r}$ $X^T \vec{r} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ -5 \\ -3 \end{bmatrix}$
 $= \begin{bmatrix} -2 & -3 & -3 \\ 0 & -5 & -3 \end{bmatrix} = \begin{bmatrix} -12 \\ -8 \end{bmatrix}$ $\nabla_{\vec{w}} J = \frac{2}{3} \begin{bmatrix} -12 \\ -8 \end{bmatrix}$ $\nabla_{\vec{w}} J = \begin{bmatrix} -8 \\ -5,3333 \end{bmatrix}$

Gradient of Bias $\frac{\partial J}{\partial b} = \frac{1}{3} (-2 -5 -3) = \frac{1}{3} (-10) = -3,3333$

Update Weights $\vec{w}^{(1)} = \vec{w}^{(0)} - \alpha \nabla_{\vec{w}} J = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0,1 \begin{bmatrix} -8 \\ -5,3333 \end{bmatrix}$
 $= \begin{bmatrix} 0,8 \\ 0,53333 \end{bmatrix}$ $\vec{w}^{(1)} = \begin{bmatrix} 0,8 \\ 0,53333 \end{bmatrix}$

Update Bias $b^{(1)} = b^{(0)} - \alpha \frac{\partial J}{\partial b} = 0 - 0,1 (-3,3333) = 0,3333$

$b^{(1)} = 0,3333$

Epoch 2

Predictions $\hat{y}^{(1)} = X \vec{w}^{(1)} + b^{(1)} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0,8 \\ 0,53333 \end{bmatrix} + \begin{bmatrix} 0,3333 \\ 0,3333 \\ 0,3333 \end{bmatrix}$

$\hat{y}_1 = 0,8 + 0,3333 = 1,1333$

$\hat{y}_2 = 2,06 + 0,53333 + 0,33333 = 2,92667$

$\hat{y}_3 = 0,83333 + 0,33333 = 1,16667$

$\hat{y}^{(1)} = \begin{bmatrix} 1,1333 \\ 2,92667 \\ 1,16667 \end{bmatrix}$

Residuals $\vec{r}^{(1)} = \hat{y}^{(1)} - y = \begin{bmatrix} 2,4667 - 2 \\ 1,8 - 5 \\ 2,2 - 3 \end{bmatrix} \quad \vec{r}^{(1)} = \begin{bmatrix} -0,5333 \\ -2,2000 \\ -0,8000 \end{bmatrix}$

Gradient of Weights $X^T \vec{r}^{(1)} = \begin{bmatrix} 2 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} -0,5333 \\ -2,2000 \\ -0,8000 \end{bmatrix}$
 $= \begin{bmatrix} -0,5333 - 4,4000 \\ -2,2000 - 0,8000 \end{bmatrix} = \begin{bmatrix} -4,9333 \\ -3,0000 \end{bmatrix}$

$\nabla_{\vec{w}} J = \frac{2}{3} \begin{bmatrix} -4,9333 \\ -3,0000 \end{bmatrix} \quad \nabla_b J = \begin{bmatrix} -3,2667 \\ -2,6667 \end{bmatrix}$

Gradient of Bias $\frac{\partial J}{\partial b} = \frac{2}{3} (-0,5333 - 2,2000 - 0,8000) = \frac{2}{3} (-3,5333)$

$\frac{\partial J}{\partial b} = -3,222$

Update Weights $\vec{w}^{(2)} = \vec{w}^{(1)} - \alpha \nabla_{\vec{w}} J = \begin{bmatrix} 0,8000 \\ 0,5333 \end{bmatrix} - 0,1 \begin{bmatrix} -4,9333 \\ -3,0000 \end{bmatrix}$
 $= \begin{bmatrix} 0,8000 + 0,4933 \\ 0,5333 + 0,3000 \end{bmatrix} \quad \boxed{\vec{w}^{(2)} = \begin{bmatrix} 1,2933 \\ 0,8333 \end{bmatrix}}$

Update Bias $b^{(2)} = b^{(1)} - \alpha \frac{\partial J}{\partial b} = 0,6667 - 0,1 (-3,222)$

$b^{(2)} = 0,9889$

(c) Cost Function Before and After Epoch 2

Cost Before Training $J^{(1)} = \frac{1}{3} [(1-2)^2 + (1-5)^2 + (1-3)^2]$

$= \frac{1}{3} (1 + 16 + 4) = 26/3 \quad J^{(1)} = 22,6667$

Cost After Epoch 2 $\vec{w}^{(2)} = \begin{bmatrix} 1,2933 \\ 0,8333 \end{bmatrix} \quad b^{(2)} = 0,9889$

Predictions $\hat{y}_1 = 1(1,2933) + 0(0,8333) + 0,9889$

$\hat{y}_1 = 2,0978$

$\hat{y}_2 = 2(1,2933) + 1(0,8333) + 0,9889$

$\hat{y}_2 = 4,0267$

$\hat{y}_3 = 1(1,2933) + 1(0,8333) + 0,9889$

$\hat{y}_3 = 3,2689$

$\hat{y}^{(2)} = \begin{bmatrix} 2,0978 \\ 4,0267 \\ 3,2689 \end{bmatrix}$

Residuals $\vec{r}^{(2)} = \begin{bmatrix} 2,0978 - 2 \\ 4,0247 - 5 \\ 1,7689 - 3 \end{bmatrix}$ $\vec{r}^{(2)} = \begin{bmatrix} 0,0978 \\ -0,9733 \\ -1,2311 \end{bmatrix}$

MSE
 $J^{(2)} = \frac{1}{3} [(0,0978)^2 + (-0,9733)^2 + (-1,2311)^2]$
 $= \frac{1}{3} [0,0096 + 0,9474 + 1,5156] = \frac{2,4726}{3}$
 $J^{(2)} \approx 0,8242$

Final Comparison

$J^{(1)} = 12,6667$ $J^{(2)} = 0,8242$

$J^{(2)} < J^{(1)}$

The cost function decreases after training